

switch the solenoid valve on and off through an electrical pulse.

Ultrasonic atomiser. It has a piezoelectric transducer that converts high-frequency electronic signal to mechanical oscillations. The oscillations create small waves in the water that get converted to micro droplets, which break surface tension of the water body and become water vapour.

Software

Install Arduino IDE in your computer and upload program *Ultrasonic_Atomizer.ino*. In the program, Arduino digital pins 12 and 13 are defined as output and its analogue pin A0 is defined as input. Main functions of the Arduino code are explained below.

pinMode(). It configures the specified pin to behave either as an input or an output. See the description of digital pins for functionality of the pins. These can also be used as PWM pins.

digitalWrite(). If the pin has been configured as an output with *pinMode()*, its voltage will be set to the corresponding value: 5V (or 3.3V on 3.3V boards) for High, 0V (ground) for Low.

Serial. It is used for communication between the Arduino board and a computer or other devices. All Arduino boards have at least one serial port (also known as a UART, or USART), and some have several ports.

begin(). Sets the data rate in bits per second (baud) for serial data transmission. For communicating with the computer, use one of these rates: 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 28800, 38400, 57600, or 115200. You can, however, specify other rates, for example, to communicate over pins 0 and 1 with a component that requires a particular baud rate.

analogRead(). Reads the value from the specified analogue pin. Arduino board contains a multichannel, 10-bit analogue-to-digital converter. This means that it will map input voltages between 0 and the operating voltage (5V or 3.3V) into integer values between 0 and 1023.

println(). Prints data to the serial port as human-readable ASCII text followed by a carriage return character.

delay(). It is used to pause the

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno board
IC1	- 7812, 12V voltage regulator
BR1	- 1A bridge rectifier
D1	- 1N4007 rectifier diode
T1	- 2N2222 NPN transistor
LED1, LED2	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 2.2-kilo-ohm
R2, R3	- 1-kilo-ohm
Capacitors:	
C1	- 1µF, 25V electrolytic
C2	- 1000µF, 35V electrolytic
Miscellaneous:	
CON1	- 2-pin connector
RL1	- 12V, single changeover relay
L1	- Ultrasonic atomiser (113kHz)
SEN18	- Water-level sensor
X1	- 230V AC primary to 15V, 2A secondary transformer
	- General-purpose PCB/Veroboard
	- Jumper wires

program for the amount of time (in milliseconds) specified as parameter.

Construction and testing

Assemble the circuit on a general-purpose PCB or Veroboard as per Fig. 1. Connect 230V AC across the primary of transformer X1.

As shown in Fig. 1, signal 'S' pin of water-level sensor SEN18 is connected with analogue pin A0 of the Arduino Uno board and its 'GND' and '+' pins are connected with ground and 5V pins, respectively. This water-level sensor will detect the water level available in the water container where the water from air-conditioner's small tray is collected.

The SPDT relay RL1 used at the output side is triggered through resistor R3, pin 13 of Arduino Uno, switching transistor T1, along with diode D1 as flywheel diode. LED2 is

connected as indicator LED with pin 12 of Arduino Uno with resistor R2 to indicate the activated/deactivated state of ultrasonic atomiser L1. The ultrasonic atomiser is connected as a load across normally-open (NO) terminal and common C terminal of relay RL1 with series connection of 12V DC power source.

There are a few things to be managed for this setup. The water level sensor is to be placed vertically in the container where the condensed water draining from the indoor unit will accumulate. And the ultrasonic atomiser disk is to be placed inside this container at its bottom.

After the setup, water level sensor should be calibrated according to the need. You can do it by following below-mentioned steps. First, upload *Calibration.ino* program in Arduino and then power on the system.

Step 1. Pour water into the water container and check the start level.

Step 2. Check the level on the serial monitor and note down the full and low-threshold levels. Here, 650 is the threshold level during testing.

After calibration, upload the *Ultrasonic_Atomizer.ino* code to Arduino board. After completing the whole setup, turn the power supply on. The water level sensor will detect the water level in the container.

When sensor detects full level of the water, Arduino will give high signal to the digital I/O pins 12 and 13 through the program. This will energise relay RL1. So, ultrasonic atomiser connected across the relay will get powered on and start atomising the water. Once water level reaches below the set level, it will automatically give low signal to digital output pin 13 of the Arduino through the program. This will de-energise relay RL1 and disconnect the power to ultrasonic atomiser L1, stopping atomising of the water.

This is how we can atomise the air-conditioner's water and eliminate emptying of buckets full of water. **EFY**

EFY Note
The source code of this project is available for free download at source.efymag.com

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